

Gas Laws

Complete By	TOPIC	PRACTICE
Section 1 Complete as Apr 28 homework	KMT Theory	
Section 2	Charles' Law Volume vs Temperature	Questions on linked sheet Charles': pg 434 #16-19
Work on Section 2 May 5- 9	Boyle's Law Pressure vs Volume	Questions on linked sheet Boyle's: pg428 #5-8
	Pressure vs Temperature (Gay Lussac's)	
Section 3 May 12 - 16	Combined Gas Law Section 3 Assignment - See slide for info	Questions at bottom of the Combined Note note
Section 4 May 20-23th	Ideal Gas Law PV=nRT	text reference p. 443-445
	Gases Textbook Solutions	Investigating Gas Laws - Virtual lab Atmospheric Gases and Health

In Class Gas Days -

Day	TOPIC	PRACTICE
Jun 6 & 9 (dates are approximate)	Ideal Gas law and Partial Pressure wrap up In class day to solidify your understanding of the Day 1-4 concepts and complete additional practice questions. Focus on partial pressures and ideal gas law Please bring all materials to utilize your time well. Practice on Partial Pressures. Answers are at the bottom	Dalton partial pressure slides Khan Partial Pressure
Jun 10	QUEST on Sections 1-3 (this will count as a small mark on final grade) After Quest: Read p 466 - 468 on combining volumes	
Jun 11	Molar Volume and Combining volumes Molar Volume Note & Slides Molar Volume Of Gas Chemical Calculations Chemistry FuseSchool Molar Volume of a Gas Lab DEMO (or Video version) (Also read text p. 469-471) Law of combining volumes (Last page of ideal gas law note, AKA Gay-Lussac's Law)	Read p 466 - 468 on combining volumes p. 471 #7-11 p. 472 #12-14 p. 474 #1-3

In Class Gas Days -

Day	TOPIC	PRACTICE
Jun 12	Value of R Lab (Class Set Handouts) Value of R lab Instructions & Image	Read stoichiometry portion of ideal law note (posted above) for next class p. 468 #1-5 finish calculations from lab
Jun 13	Gas Stoichiometry Read end of ideal gas law sheet and work on questions for homework. Gas Law Stoichiometry Worksheet Practice.pdf - Answers Note that the answer sheet uses a different problem solving method but you should be able to match your calculation and certainly see if you get the correct answer.	p. 483 #1-4 p. 486 #1-4

Teacher Note f

IMPORTANT NOTES

SCH3U Terms & Translations

Gas Laws Summary

The above summary is great. A few notes though:

We will mostly use the value of R of:

8.31 L•kPa•mol⁻¹•K⁻¹. This R value is in kPa which is why the number is different.

ALSO: the current international Standards are different than your textbook and some of the notes here. However, we will use the same as your textbook:

STP: 0C (273K) and P = 101.3 kPa

SATP: 25C (298K) and P = 100kPa

Read Kinetic molecular theory p109-110 and p419-420. Make own notes on it.

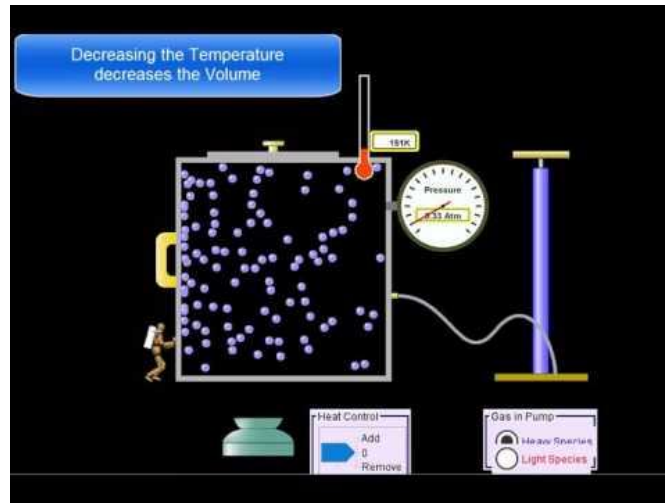
[KMT Part 1 Video](#)

[KMT Part 2 Video](#)

[Absolute zero video](#)

[Gas Particles Note & Questions](#) is another optional reading & practice

Volume vs Temperature



Charles' Law ex. 1

A 1.00 L sample of H_2S gas, at 20°C , is heated until the temperature reaches 40°C . What is the new volume of the gas?

$$\frac{V_1}{V_2} = \frac{T_1}{T_2} \quad 1.00\text{L} =$$

$$T_1 = 20 + 273 = 293 \text{ K}$$

$$T_2 = 40 + 273 = 313 \text{ K}$$

Charles' Law ex. 2

When 5.0 L of $\text{CO}(\text{g})$ at a temperature of 50°C is transferred to a 2.5 L flask, the temperature decreases. What is the temperature at this new volume?

$$\frac{V_1}{V_2} = \frac{T_1}{T_2} \quad \frac{5.0}{2.5} = \frac{323}{T_2}$$

$$V_1 = 5.0 \text{ L} \quad T_1 = 50^\circ\text{C} + 273 = 323 \text{ K}$$

$$V_2 = 2.5 \text{ L} \quad T_2 = ?$$

$$T_2 =$$

Charles' Law ex. 3

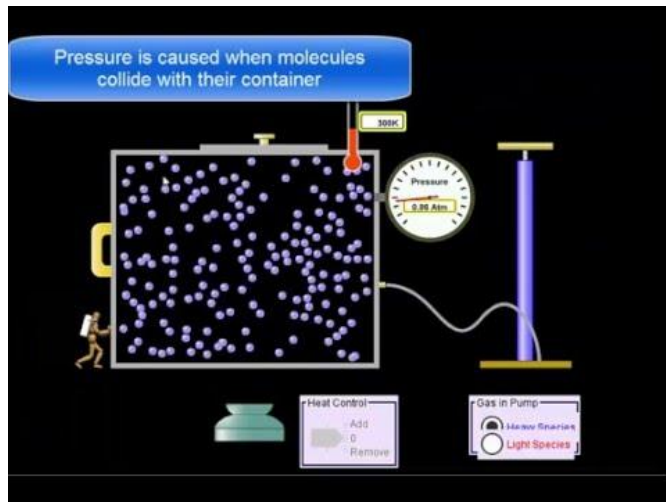
When 0.75 mL of $\text{N}_2(\text{g})$, at 20°C , is heated, its volume doubles. What is the temperature at this new volume?

$$\frac{V_1}{V_2} = \frac{T_1}{T_2}$$

$$V_1 = 0.75 \text{ mL} \quad T_1 = 20^\circ\text{C} + 273 = 293 \text{ K}$$

$$V_2 = 1.50 \text{ mL} \quad T_2 = ?$$

Pressure vs. Volume



Boyle's Law ex. 1

A sample of Hydrogen gas is at 1.0 atm and a volume of 22.4 Liters (L). What pressure would the gas have if the volume dropped to 11.2 L?

$$\frac{P_1}{P_2} = \frac{V_2}{V_1}$$

Boyle's Law ex. 2

A sample of CO₂ gas has a pressure of 5.65 atm in a 125 mL flask. The sample is transferred to a new flask where it has a pressure of 6.99 atm at the same temperature. What is the volume of the new flask?

$$\frac{P_1}{P_2} = \frac{V_2}{V_1}$$

$$\frac{5.65}{6.99} = \frac{V_2}{125}$$

$$P_1 = 5.65 \text{ atm} \quad V_1 = 125 \text{ mL}$$

$$P_2 = 6.99 \text{ atm} \quad V_2 = ?$$

Boyle's Law ex. 3

What would you expect the volume of a 3.00 L sample of He to be if its pressure is increased from 1.50 atm to 2.0 atm?

$$\frac{P_1}{P_2} = \frac{V_2}{V_1}$$

$$\frac{1.50}{2.0} = \frac{V_2}{3.00}$$

$$P_1 = 1.50 \text{ atm} \quad V_1 = 3.00 \text{ L}$$

$$P_2 = 2.0 \text{ atm} \quad V_2 = ?$$

Continued Next Slide

Pressure and Volume of Gases Sheet

[Gas Pressure Basics Video](#)

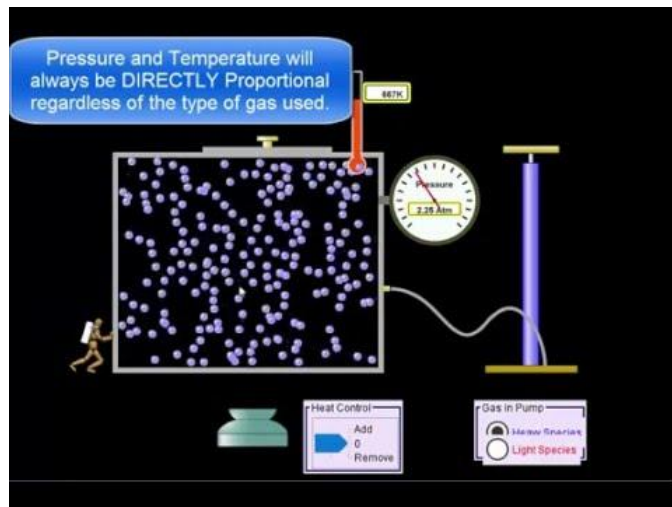
[Pressure Conversions Video](#)

[Pressure Animation](#)

[Boyle's Law Video](#)

Pressure vs. Volume

Pressure vs Temperature



P vs T Ex. 1

Ar(g) is sealed in a metal canister at a pressure of 1.0 atm and a temperature of 20°C. Over the next hour, the temperature is raised to 50°C. What will be the resulting pressure after this time?

$$\frac{P_1}{P_2} = \frac{T_1}{T_2}$$

$$\frac{1.0}{P_2} = \frac{293}{323}$$

$$P_2 = ?$$

P vs T Ex. 2

Alex is given a Helium filled balloon when he goes to Chuck E Cheese's. When he goes outside he starts crying because the balloon shrinks. If the balloon started at a pressure of 1.0 atm and 20°C, and the temperature outside is 5°C, what is the pressure in the balloon (ignoring changes in volume)?

$$\frac{P_1}{P_2} = \frac{T_1}{T_2}$$

$$\frac{1.0}{P_2} = \frac{293}{278}$$

$$P_2 = ?$$

Self-Check Quiz

Answers

Combined Gas Law

** FYI Some online materials may use different STP values & you may see different numbers in university depending on your program.

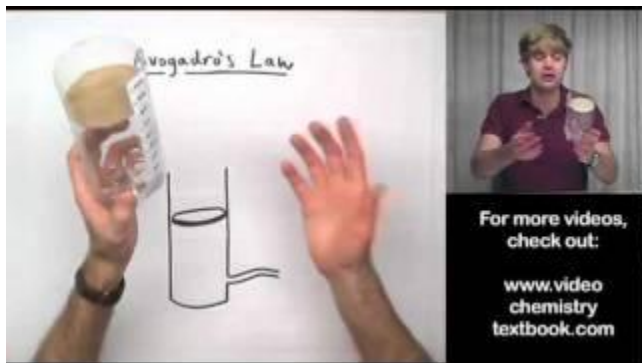
(Try to fill in the blanks by yourself, for help see following video)

Rearranging the Combined Gas Law Video

Section 3 Assignment: SUBMIT Questions #5-7 from The Combined Note. FULL solutions must be clearly shown.

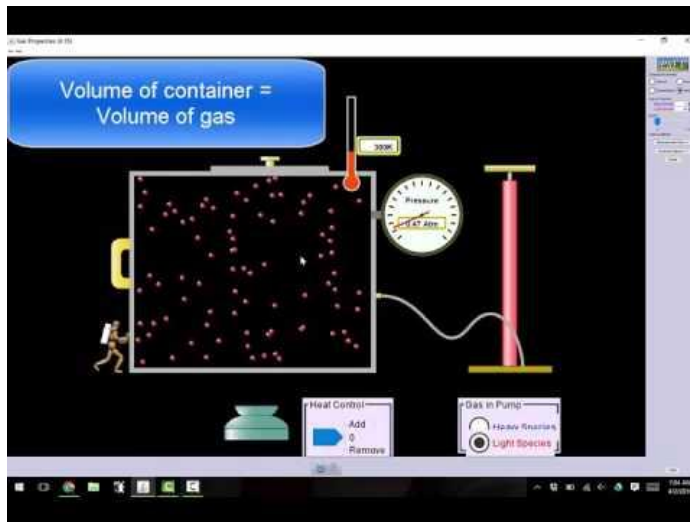
Avogadro's Hypothesis

Avogadro's law states that "equal volumes of all gases, at the same temperature and pressure, have the same number of molecules."



Continue to next slide for examples

Ideal Gas Law Note



Introduction & $PV=nRT$

E. Ideal Gas Law

- Ideal Gas—gas that behaves perfectly under all situations

$$PV = nRT$$

Diagram showing the units for each variable in the Ideal Gas Law equation:

- P (Pressure) in atm
- V (Volume) in L
- n (moles)
- R (Gas Constant) in 0.0821
- T (Temperature) in K

STP—Standard Temperature and Pressure.
 $T = 0.0^\circ\text{C}$ and $P = 1.0\text{ atm}$

NOTE ON VALUE OF R:

Most of our questions will use **$8.31 \text{ L}\cdot\text{kPa}\cdot\text{mol}^{-1}$**

• **K^{-1}** . The numerical value of R can be different if the units, especially for pressure are different

1 set of conditions **Ideal Gas Law Ex. 1**

A 1.25g sample of CO_2 is contained in a 750.0 mL flask at 22.5°C . What is the pressure of the gas?

$PV = nRT$

$P = ?$

$V = \frac{750.0 \text{ mL}}{1000} = 0.7500 \text{ L}$

$n = \frac{1.25 \text{ g CO}_2}{44.01 \text{ g/mol}} = 0.0284 \text{ mol}$

$T = 22.5 + 273 = 295.5 \text{ K}$

$P(0.7500) = (0.0284)(0.0821)(295.5)$

Ideal Gas Law Ex. 2

What volume will 50.0 g of Ammonia gas, NH_3 , occupy at a temperature of 0.0°C and a pressure of 1.0 atm? $\xrightarrow{\text{STP}}$

$PV = nRT$

$P = 1.0 \text{ atm}$

$V = ?$

$n = \frac{50.0 \text{ g NH}_3}{17.04 \text{ g/mol}} = 2.93 \text{ mol}$

$T = 0 + 273 = 273 \text{ K}$

$(1.0)V = (?)$

[Ideal Gas Law Problems Video](#)

Converting between [mole & volume at STP](#)

[How to avoid memorizing ALL the gas laws](#)

Video [Choosing the RIGHT Gas Equation](#)